SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Lecture Series #09 - Digital Comparators

Example 01 Video Link - https://youtu.be/YsixHklZI1s

Design 2-bit Comparator using gates

The truth table for 2-bit Comparator is

Inputs				Outputs		
A1	A0	B1	B0	A > B	A = B	A < B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

A > B

A1 A0 \ B1 B0	00	01	11	10
00				
01	1			
11	1	1		1
10	1	1		

$$A>B = A_0 \overline{B_1}\,\overline{B_0} + A_1 \overline{B_1} + A_1 A_0 \overline{B_0}$$

A = B

A_1A_0 \ B_1B_0	00	01	11	10
00	1			
01		1		
11			1	
10				1

A < B

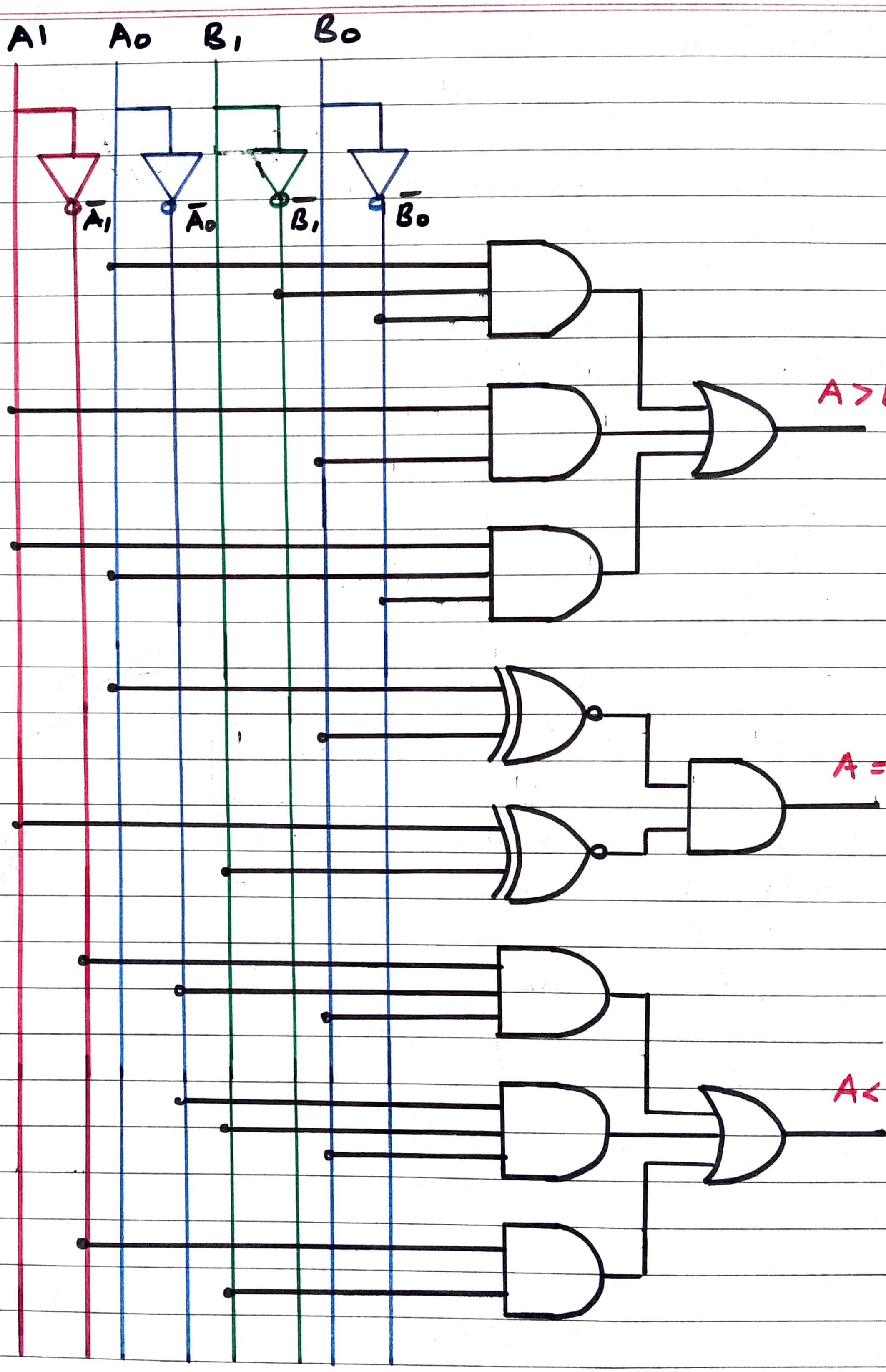
A_1A_0 \ B_1B_0	00	01	11	10
00		1	1	1
01			1	1
11				
10			1	

$$(A<B) = \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0 + \bar{A}_1 B_1$$

$$(A = B) = \bar{A}_1 \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}_1 A_0 \bar{B}_1 B_0 + A_1 A_0 B_1 B_0 + A_1 \bar{A}_0 B_1 \bar{B}_0$$

$$= \bar{A}_1 \bar{B}_1 (\bar{A}_0 \bar{B}_0 + A_0 B_0) + A_1 B_1 (A_0 B_0 + \bar{A}_0 \bar{B}_0)$$

$$= (A_0 \odot B_0)(A_1 \odot B_1)$$


A1
A0
B1
B0
$\bar{A}_1$
$\bar{A}_0$
$\bar{B}_1$
$\bar{B}_0$
A>B
A=B
A<B

Example 02

Video Link - https://youtu.be/Glu27lyXI_s

Design a 1 bit Comparator using logic gates

Consider a two one bit number A and B.
The truth table is shown below

Inputs		Output Y		
A	B	A=B	A>B	A<B
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

A=B

A \ B	0	1
0	1	
1		1

A>B

A \ B	0	1
0		
1	1	

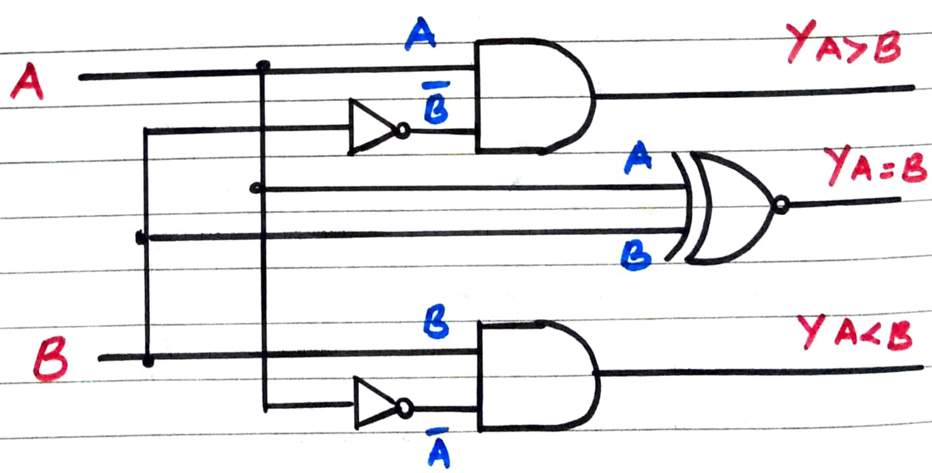
A<B

A \ B	0	1
0		1
1		

$Y(A=B)$
$= \bar{A}\bar{B} + AB$
$= \overline{A \oplus B}$
$= A \odot B$

$Y_{A>B}$
$= A\bar{B}$

$Y_{A<B}$
$= \bar{A}B$



Video Link - https://youtu.be/m3HwoWaBV_Q

Example 03

Design a 4-bit magnitude Comparator

Comparison Algorithm

$A = B$

→ Consider two 4-bit binary numbers A and B such that

$A = A_3 A_2 A_1 A_0$ and $B = B_3 B_2 B_1 B_0$

→ Two numbers are said to be equal when $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$ and $A_0 = B_0$

→ In binary, the equality of each pair of bits can be expressed logically with an enclusive NOR function as

$$x_i = A_i B_i + \bar{A_i} \bar{B_i} \quad \text{for } i = 0, 1, 2, 3$$

where $x_i = 1$ only if the pair of bits in position i are equal (ie if both are 1 or 0)

→ Therefore we can say that

$$A = B, \text{ if } x_0 \cdot x_1 \cdot x_2 \cdot x_3 = 1$$

$(A > B)$ or $(A < B)$

→ To determine whether A is greater or less than B, it is necessary to inspect the relative magnitudes of pairs of significant digits, starting from the most significant position

→ If the two digits of a pair are equal, we compare the next lower significant pair of digits. The comparison continues until a pair of unequal digits is reached

→ If Corresponding digits of A is 1 and that of B is 0, we say that A > B

→ If Corresponding digits of A is 0 and that of B is 1, we say that A < B

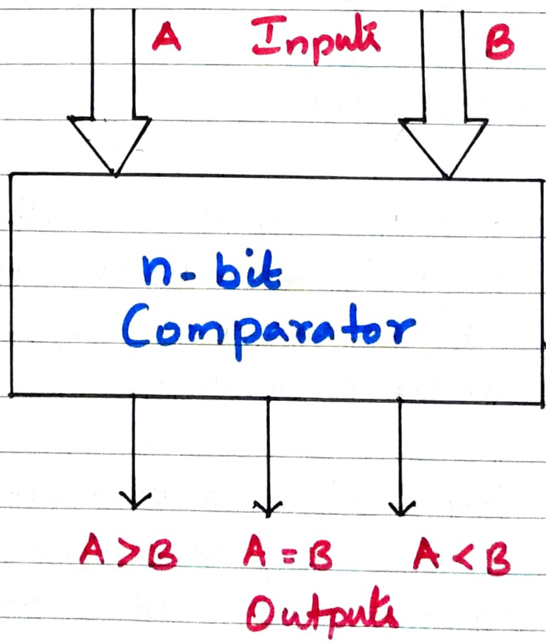
→ The sequential expression to determine whether A > B or A<B are as given below

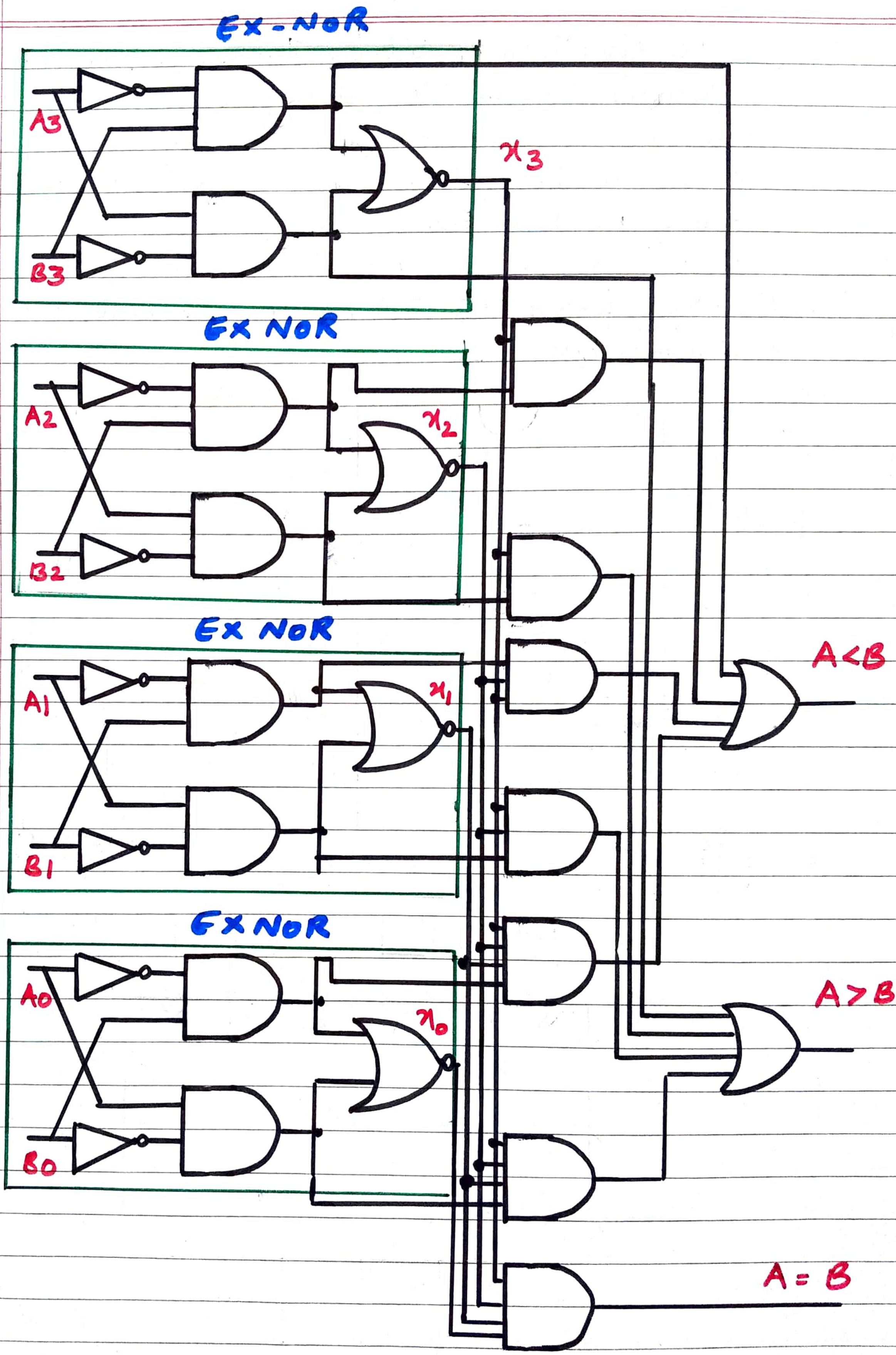
$$(A > B) = A_3 \overline{B_3} + x_3 A_2 \overline{B_2} + x_3 x_2 A_1 \overline{B_1} + x_3 x_2 x_1 A_0 \overline{B_0}$$

$$(A < B) = \overline{A_3} B_3 + x_3 \overline{A_2} B_2 + x_3 x_2 \overline{A_1} B_1 + x_3 x_2 x_1 \overline{A_0} B_0$$

The block diagram of an n-bit Comparator is shown below. It receives two n-bit numbers A and B as inputs and the outputs are A > B, A = B and A < B.

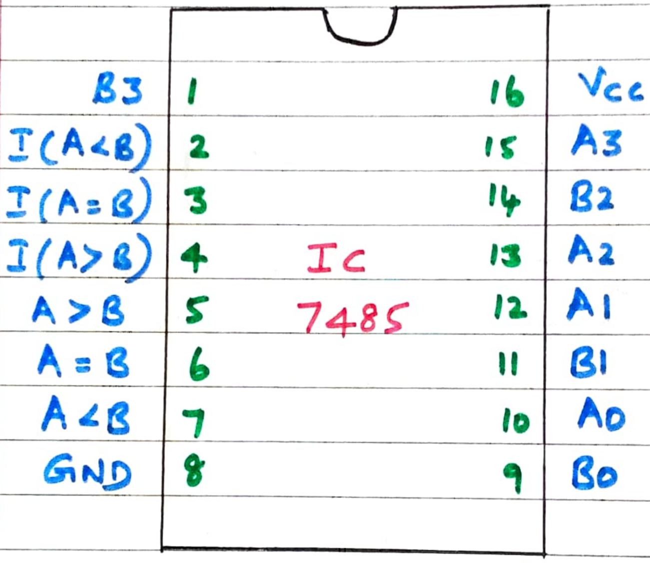
Depending upon the relative magnitudes of the two number, one of the outputs will be high.




EX-NOR
A3
B3
x3
EX NOR
A2
B2
x2
EX NOR
A1
B1
x1
EX NOR
A0
B0
x0
A<B
A>B
A=B

Video Link - https://youtu.be/7l2KifP2_J4

IC 7485 is a 4-bit Comparator. It Can be Used to Compare two 4-bit binary words by grounding I(A<B), and I(A>B) and Connecting input I(A=B) to Vcc. These IC's Can be Cascaded to Compare words of almost any digits lengths. Its 4-bit inputs are weighted ($A_0 - A_3$) and ($B_0 - B_3$), where A_3 and B_3 are the most Significant bits



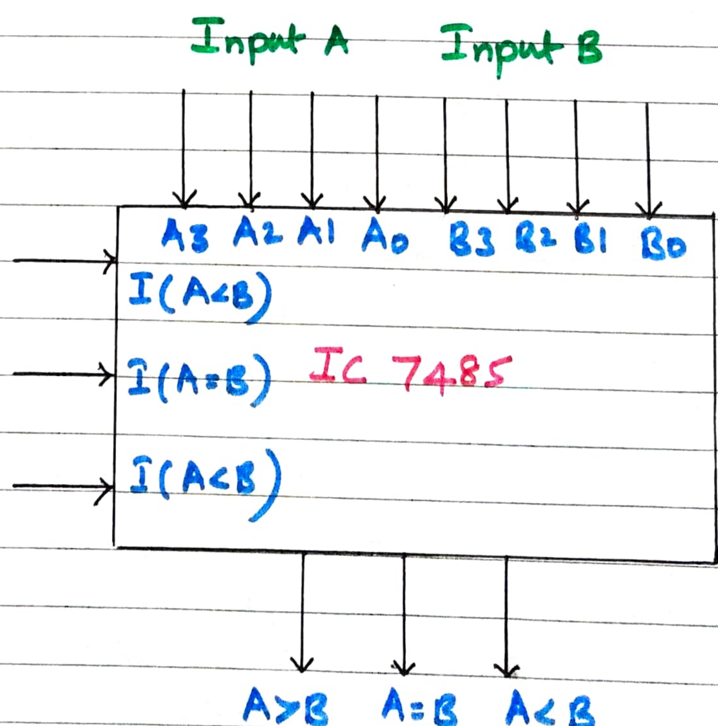


Diagram shows the logic symbol and pin diagram of IC 7485.

The operation of IC 7485 is described in the function table, showing all possible logic conditions.

Comparing Inputs A B	Cascading Inputs			Outputs		
	I(A>B)	I(A=B)	I(A<B)	A>B	A=B	A<B
A > B	X	X	X	1	0	0
A = B	1	0	0	1	0	0
	X	1	X	0	1	0
	0	0	1	0	0	1
	0	0	0	1	0	1
	1	0	1	0	0	0
A < B	X	X	X	0	0	1